Figure 2.1 The surface of the water starts out flat and remains so as the bucket starts to spin. Subsequently, as the water also starts to spin, its surface becomes concave, and it remains concave while the water spins, even as the bucket slows and stops.

Relativity Before Einstein

"Relativity" is a word we associate with Einstein, but the concept goes much further back. Galileo, Newton, and many others were well aware that *velocity*—the speed and direction of an object's motion—is relative. In modern terms, from the batter's point of view, a well-pitched fastball might be approaching at 100 miles per hour. From the baseball's point of view, it's the *batter* who is approaching at 100 miles per hour. Both descriptions are accurate; it's just a matter of perspective. Motion has meaning only in a relational sense: An object's velocity can be specified only in relation to that of another object. You've probably experienced this. When the train you are on is next to another and you see relative motion, you can't immediately tell which train is actually moving on the tracks. Galileo described this effect using the transport of his day, boats. Drop a coin on a smoothly sailing ship, Galileo said, and it will hit your foot just as it would on dry land. From your perspective, you are justified in declaring that you are stationary and it's the water that is rushing by the ship's hull. And since from this point of view you are not moving, the coin's motion relative to your foot will be exactly what it would have been before you embarked.

Of course, there are circumstances under which your motion seems intrinsic, when you can feel it and you seem able to declare, without recourse to external comparisons, that you are definitely moving. This is the case with *accelerated* motion, motion in which your speed and/or your direction changes. If the boat you are on suddenly lurches one way or another, or slows down or speeds up, or changes direction by rounding a bend, or gets caught in a whirlpool and spins around and around, you know that you are moving. And you realize this without looking out and comparing your motion with some chosen point of reference. Even if your eyes are closed, you know you're moving, because you feel it. Thus, while you can't feel motion with constant speed that heads in an